Benjamin Pomeranz

201-655-5094 | benjamin_pomeranz@brown.edu | benpomeranz.com | github.com/benpomeranz

Education

Brown University — Providence, RI
Pursuing Sc.B. in Mathematics and A.B. in Computer Science (for May 2026) — *Sep. 2022 – Present*

- **Math**: Graduate real function theory, manifolds, real and complex analysis, number theory, cryptography, linear and abstract algebra, graph theory, differential geometry, modern applications of probability, honors statistics
- **Computer Science** Deep Learning (Implementing various neural nets in TF and Pytorch), Graduate Algorithms (Linear Programming, Graph Sparsification, Multiplicative Weights), Cryptography, Computer Systems, etc.
- Building a strong skillset in nonfiction writing
- 4.0 GPA

Budapest Semester in Mathematics — Budapest, Hungary
Intensive study abroad program in math — *Jan. – May 2025*

- Studied Galois theory, Benjamini-Schramm convergence of graphs, algebraic topology and "Conjecture and Proof"
- Researched the 3 dimensional 2 player subtraction game with Istvan Miklos
- Highest honors (awarded to <15% of students in the program)

Employment and Research

REU in Computational Mathematics at Montana State University — Bozeman, MT
Researcher in metabolic models — *June 2025 – Present*

- Developed and implemented algorithm in Python to 2x speed of minimal EFM decomposition for flux vectors and enable numerical stability on genome scale models for first time
- Leveraged convex geometry of the flux cone and linear programming to prove efficacy of algorithm
- Collected and analyzed data on algorithmic performance using Pandas, NumPy, and SciPy
- Presented research at JMM, and currently completing transcript to be published with Dr. Bree Cummins

Aqua Voice, Summer Intern — San Francisco, CA
Intern — *June – August 2024*

- Implemented new windowing algorithm in Aqua's Python backend, improving latency by ∼3x
- Cleaned datasets and finetuned new LLMs to reduce dictation errors on testset by ∼1.5x
- Automated model finetuning and evaluation processes
- Integrated an open source voice activity detection model, improving accuracy and latency
- Implemented latency metrics stored in PostgreSQL database

Brown University CS Department — Providence, RI
TA for CS200: Data Structures and Algorithms I — *January-May 2024*

- Teaching students arithmetic computation, sorting, BSTs, hash tables, graph algorithms, and DP
- Running labs and hours with up to 30 students

Brown University Athletics Department, Student Athlete Tutor — Providence, RI
Tutored collegiate student athletes at Brown University — *October – December 2023*

- Tutored in Math 0100 (Calculus II) and Math 0540 (Linear Algebra with theory)

Organizational Experience

Brown Effective Altruism — Providence, RI
Copresident — *June 2025 – Present*

- Leading intro fellowship reading group on existential risks, global health, and animal welfare
- Helping organize and designing website and data management for EA Northeast Retreat 2026
- Organizing monthly speaker presentations, monthly socials, and joint events with other clubs

Tenafly High School — Tenafly, NJ
Student Organization President — *Sep. 2021 – June 2022*

Skills and miscellany

Languages: Python, SQL **Rusty Languages**: Java, C
Programming Interfaces and Tools: TensorFlow, Pytorch, Keras, Git, Unix, VSCode, IntelliJ, GDB, NumPy
General Productivity: LaTeX, Google Suite, Linear, Notion, written communication, public speaking
Hobbies: Running, hiking, climbing, ultimate frisbee, crosswords, reading